

WiP: SAFEC: Slack-Aware Forward Error Correction

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Abstract

Modulation and Coding Scheme (MCS) tables expose only discrete, jointly-coupled operating points along the throughput-reliability continuum, with an average of 19.4% of achievable throughput between adjacent steps. We present Slack-Aware Forward Error Correction (SAFEC), a method that reclaims unused Forward Error Correction (FEC) correction capacity to provide fine-grained interpolation between MCS steps. SAFEC encodes a secondary data stream into the positions of deliberately-inserted, recoverable errors within a base FEC codeword, requiring no protocol redesign or new hardware. We demonstrate SAFEC's feasibility on LDPC-coded OFDM transmissions using software-defined radios, showing throughput gains of up to 9.5% over the base protocol.

CCS Concepts

• **Networks** → **Network protocol design**; • **Security and privacy** → *Security protocols*.

Keywords

sub-protocol, error correction codes

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1 Introduction

By varying both the modulation scheme and forward error correction (FEC) applied to data, most modern communication protocols are able to tune the tradeoff between throughput and reliability. Protocols like WiFi, LTE, and DVB-S2 all

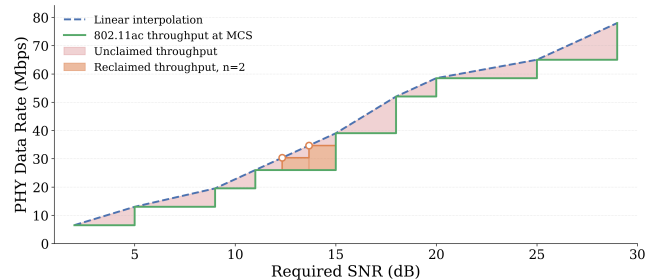


Figure 1: Throughput gains achievable by inserting $n = 2$ points between adjacent MCS values, an average of 12.9% overall.

have well-defined Modulation and Coding Scheme (MCS) tables that allow a device to choose points along this throughput-reliability continuum. However, the discrete pairs of modulation and coding rates result in significant loss between adjacent steps. For example, the 802.11ac MCS table has an average step size of 19.4% of the current MCS rate [5]. As shown in Figure 1, if even two extra points were interpolated into this table, an average 12.9% overall increase in throughput could be achieved.

To allow protocols more fine-grained control of their throughput while maintaining consistency with current MCS standards, we present Slack-Aware Forward Error Correction (SAFEC), a method of providing fine-grained interpolation of effective code rates. While the modulation schemes used are fixed to discrete steps by nature, FEC codes are highly configurable, allowing redundancy to be controlled in fine increments. Most MCS tables do not take advantage of this however, as their FEC codes are designed for worst-case conditions. This means that real channel conditions often produce many fewer errors than a code's correction capacity is provisioned for. SAFEC re-purposes a controlled portion of the unused correction budget as additional transmission capacity, allowing devices to more precisely tune their throughput. Because SAFEC operates on the codeword structure common to any block FEC code, it generalizes across protocols and can be deployed on existing hardware, improving performance without requiring architectural changes.



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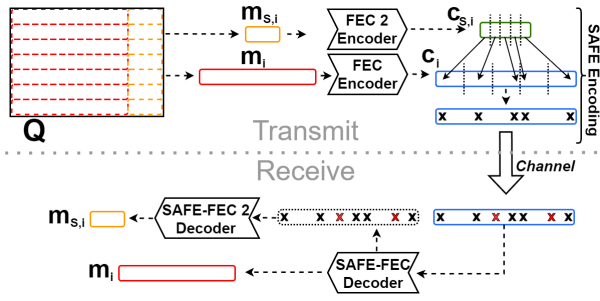


Figure 2: A visual depiction of the SAFEC encoding and decoding method.

2 Related Work

While sub-channels within error-correcting codes have been researched before, such studies focused on error insertion as a security vulnerability or a covert channel. Marzin et al. construct covert channels by embedding messages within minimal-weight codewords of error-correcting codes, allowing a receiver to recover hidden data while preserving the carrier transmission [3]. Harley et al. embed covert data within the parity bits of an LDPC codeword in 802.11ad, leveraging link adaptation and MCS selection to increase covert channel throughput, at the cost of increased uncorrectable packet errors or reduced system throughput [2]. Subramani et al. demonstrate a hardware Trojan attack that exploits surplus FEC correction margin to leak information, and propose a Trojan-agnostic detection method that identifies such attacks via inconsistencies in the receiver's noise distribution [4].

While these papers present novel contributions regarding potential uses of extra FEC capacity, our work expands the idea of error insertion beyond the scope of security and covert channels, applying it instead as a legitimate, protocol-level performance mechanism. Because SAFEC does not need to remain covert, it can take better advantage of the available FEC capacity, allowing for greater generalization and performance gains. Unlike prior fixed-insertion schemes, SAFEC encodes multi-bit symbols within the available correction budget, increasing the achievable data rate per unit of reclaimed capacity. The theory behind SAFEC coding is also applicable to previously written works, whether covert or not. Our research goal is to create an end-to-end system that demonstrates the capabilities of a SAFEC encoder across multiple protocols.

3 System Architecture

3.1 A Method for Error Insertion

The baseline operation of a protocol is to partition some large data Q into several messages $(m_i)_{i=0}^N$ of length k which are

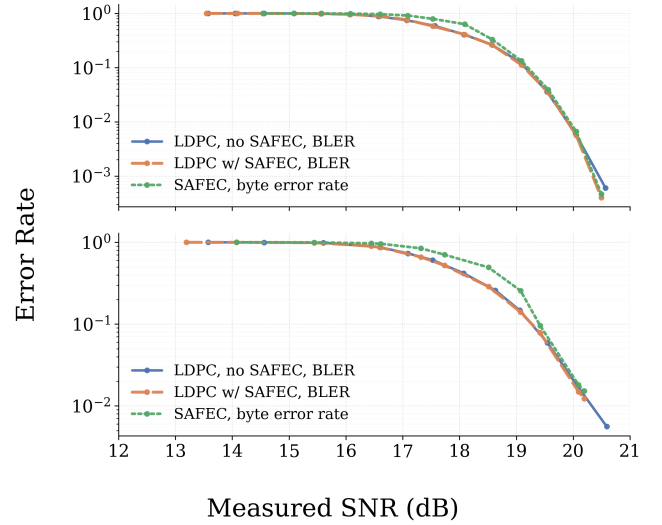


Figure 3: Initial testing results showing LDPC block error rate (BLER) and Hamming byte error rate across various SNR values, both in simulation (top) and in OTA testing (bottom).

then transmitted across a channel. SAFEC encoding divides Q into two groups: a set of messages $(m_i)_{i=0}^M$ of length k , and a set of messages $(m_{s,i})_{i=0}^M$ of length k_s . After they have been FEC encoded by their respective encoders (m_i by the base protocol's, $m_{s,i}$ by the SAFEC encoder's), the resulting set of codewords $(c_{s,i})_{i=0}^M$ are embedded as errors into $(c_i)_{i=0}^M$, allowing the same data Q to be carried in $M < N$ messages, increasing the protocol's effective throughput without altering the underlying channel conditions.

Figure 2 demonstrates how the data Q is decomposed, encoded, and transmitted, as well as showing the reverse actions done by the receiver to acquire both messages. To embed $c_{s,i}$ into c_i , a preselected value δ is used to divide c_i into segments, each of which holds a single bit error whose index in the segment encodes data from $c_{s,i}$.

This new set of codewords is then transmitted, where the channel introduces some additional errors e . For each codeword received, as long as $\text{wt}(e) < t - \delta$, the original set of codewords $(c_i)_{i=0}^M$ are recoverable. In like manner, as long as $\text{wt}(e) < t_s$, the correction capacity of the FEC used by the SAFEC encoder, the indexes of the errors found in c_i can be used to recover $c_{s,i}$.

3.2 Hardware Implementation

We implement SAFEC in GNU Radio Companion, using LDPC(100, 42) encoded OFDM blocks as the base protocol, and Hamming(7,4) encoding for the SAFEC message inserted

on top, with $\delta = 1$. With 42 bits of data from the base protocol per block and 4 bits of SAFEC data inserted per block, SAFEC increases the overall system throughput by 9.5%.

To identify the error locations, we first store the original bits received, then decode the received codeword. We then re-encode the decoded payload and compare it with the originally received bits. The XOR of these two packets indicates the locations of both the inserted errors and the natural errors from the channel. If two errors exist in a single segment, we choose the leftmost bit. We then attempt to decode the Hamming(7,4) codeword to get the SAFEC bits.

To study the effects of SAFEC on the base LDPC code, we run tests both in simulation across an additive white Gaussian noise (AWGN) channel, and over-the-air (OTA) using two USRP B210s to demonstrate its feasibility. Our results, shown in Figure 3, demonstrate that across a sample of 180k and 190k blocks per condition, the SNR required to achieve a given LDPC codeword success rate with SAFEC insertions was within measurement resolution (~ 0.1 dB) of the SNR required without SAFEC insertions, both in simulation and over the air. This highlights SAFEC's ability to effectively increase throughput without threatening reliability.

4 Future Work

4.1 Sub-Protocol Design

One challenging problem is to design a mechanism that can properly choose the value δ to balance the insertion rate with the SNR of the channel and the capacity t of the primary FEC code. Currently, we are looking into a feedback loop like Outer-Loop Link Adaptation (OLLA) [1], a control mechanism used in 5G NR that uses ACK/NACK patterns to determine the success rate of the active MCS.

Another area of research is to reduce the required metadata needed to support the SAFEC codeword, further improving SAFEC's goodput. We have developed a method to calculate the sizes and placements of the segments based on a pre-established function of δ , and future work will create a method for the TX and RX to agree on what δ was implicitly.

A related challenge arises when natural errors land in SAFEC segments, forcing a receiver to decide which error was SAFEC encoding. While initial tests blindly guess the leftmost bit, an algorithm that could evaluate the errors to determine which was most likely to have been inserted could eliminate the need for a secondary FEC code entirely. The ability to accurately quantify which errors were caused by the channel could also allow SAFEC to assist in recovering messages whose error count exceeds the error capacity of the base FEC code.

As the sub-protocol develops, we hope to extend its adaptability to convolutional codes, allowing SAFEC encoding to be used on virtually any base protocol. For example, while

WiFi's Very High Throughput (VHT) and 802.11ax modes uses LDPC codes, the baseline FEC codes used on WiFi are convolutional codes that lack a fixed block structure, making error-location extraction an open problem.

4.2 Physical Implementations

With the exciting results found in our initial testing on the B210 SDRs, we are now seeking to fine-tune our setup so SAFEC may be applied to full protocol stacks such as WiFi 6 and 5G NR. As we continue to develop the requirements for this sub-protocol, we will also test on hardware to validate the system's emerging capabilities. We will optimize the SDR test bench to become a platform for measuring the results of each SAFEC iteration. This will require refining many different parameters we have held constant thus far, such as using only $\delta = 1$ in testing and holding channel conditions constant. Furthermore, we do not wish to keep SAFEC relegated to the world of SDRs. To prove its usefulness in other cases, we hope to implement it on different Commercial Off-The-Shelf (COTS) products. At the time of writing, we have plans to make working systems in Simulink and OpenWifi while also branching out to different SDRs, with an end goal of creating a software package that could run on a laptop or cell phone.

SAFEC provides a way of opportunistically reclaiming unused data rate, demonstrating a promising path forward to improve the overall adaptability of modern wireless devices by tapping into this unused capacity.

Acknowledgments

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